

Oscar Fernandez

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Experience

Backend Developer & DevOps

April 2026 — Present

Intake · Lanzadera, Valencia

B2B SaaS platform for customer support and feedback management powered by AI, featuring an advanced conversational agent, knowledge base and ticketing system.

- Design and full ownership of the AWS infrastructure for the MVP: ECS Fargate, ECR, ElastiCache Redis, S3 + CloudFront for frontends and Secrets Manager; stack selection and architecture decisions made independently.
- Implementation of the CI/CD pipeline with GitHub Actions: `main` → pre-production flow / `v*` tags → production with manual approval, covering 10 ECS services across 4 repositories.
- Development of the core backend in FastAPI: domain error system, rate limiting, validation and observability patterns for production.
- Built the integrations with **Slack**, **Microsoft Teams** and **WhatsApp Business** for the conversational agent.
- Design and implementation of the RAG pipeline: embeddings, retrieval and prompt construction, with final inference on **Claude (Amazon Bedrock)**; function calling and custom tools system for agent orchestration.

Python · FastAPI · MongoDB · PostgreSQL · Redis · AWS (ECS, ECR, S3, CloudFront, ElastiCache, Secrets Manager, Amazon Bedrock) · Docker · GitHub Actions

Full Stack Developer Intern

September 2025 — April 2026

MooDest · Mataró

B2B SaaS AI platform for HR: intelligent helpdesk with conversational agent and people analytics.

- Development of REST API with FastAPI for integration with a time-tracking platform of **300,000+ users**: authentication, role management, KPIs and security.
- Built **4 integrations** with external channels (Slack, Teams, among others) for the conversational agent, from protocol design to deployment.
- Lead developer of the people analytics dashboard (Angular + TypeScript + NestJS): feature implementation, complex user flows and end-to-end business logic.
- DevOps: containerization with Docker Compose, Nginx configuration in production and deployment management on self-hosted server.
- Agile methodology with Scrum/Kanban, code reviews and PR management in a team environment.

TypeScript · Angular · NestJS · Python · FastAPI · MongoDB · Redis · Docker · Nginx

Relevant Personal Projects

- **ArdiQ** (Rust / Python / PyO3 / Tokio) [active]
Python task queue with a Rust core. ARQ-compatible API, async runtime built on Tokio, native bindings via PyO3/maturin. Published on PyPI as ardiq. [github](#) →
- **Fluxo** (FastAPI / React / PostgreSQL / Redis) [active]
Decentralized crypto payment platform à la Stripe with multi-chain support (Ethereum, Solana, Polygon) and two custody models for merchants. [github](#) →
- **Argus** (FastAPI / React / PostgreSQL / Redis)
OSINT platform for real-time monitoring of Telegram bots and chats. Multi-bot manager, Redis Pub/Sub + SSE for live streaming, JWT auth and dockerized deployment. [github](#) →
- **shellox** (Rust)
Encrypted reverse shell with multi-session support. Encrypted communications and concurrent connection management with Tokio. [github](#) →

Technical Skills

Languages	Python, TypeScript/JavaScript, Rust, Bash, SQL
Backend	FastAPI, NestJS, Django, ARQ, Celery, Celery Beat, aiohttp, asyncio
Frontend	Angular, React, Vite, TanStack Query, shadcn/ui
Databases	MongoDB Atlas, PostgreSQL, Redis (cache, pub/sub, rate limiting, queues)
Cloud & DevOps	AWS (ECS Fargate, ECR, S3, CloudFront, ElastiCache, Secrets Manager, SageMaker) Docker, Docker Compose, GitHub Actions, Nginx, Linux
AI / ML	RAG pipelines, Ollama, Hugging Face, Anthropic API, Amazon Bedrock, SSE streaming
Security	Pentesting, C2, reverse shells, AV evasion, OSINT, Hack The Box (25+ machines)
Protocols	REST, WebSockets, SSE, Pub/Sub, JWT, OAuth2
Tools	Git, Telethon, aiogram, structlog, SQLAlchemy, Sentry, pnpm, uv

Education

Project-driven self-taught learning	2020 — Present
Most of my stack has been acquired independently, reading official documentation, books, building real projects from scratch and solving problems in production. Every technology listed on this CV has been used in a real context, not just in tutorials.	
Vocational Diploma in Computer Systems and Networks (SMR)	2024 — 2026
Escola Pia Santa Anna, Mataró — Spanish Vocational Education	
Web Development and Programming Diploma	2024 — 2026
Frontend (HTML, CSS, JS, React, NextJS) and backend (Python: Django, Flask, FastAPI)	
Compulsory Secondary Education (ESO)	2020 — 2024
Institut Esteve Albert, Sant Vicenç De Montalt	

Additional Training

- **Hack4u (Cybersecurity Academy)**: Offensive Python, Introduction to Hacking, Introduction to Linux (2024)
- **Rust Programming** (2025): ownership, borrowing, concurrency with Tokio, systems programming and personal projects published.
- **Personal portfolio** (bytay.dev): personal website with portfolio, projects, CV and a technical blog with articles on pentesting, malware, C2 and backend development.